Grassland and Dairy Production

Land Sciences

May, 2026

Grassland and Dairy Production (A11414)

Short Title: Grassland and Dairy Production
Faculty Science and Computing
Department Land Sciences
Cluster Agriculture
Author JRYAN
Credits 5 **Level:** Introductory

Description of Module / Aims

This module will enable the student to gain a comprehensive understanding of dairy production and grassland management. It will provide students with the skills and knowledge to manage a dairy herd, rear replacement heifers and produce quality milk in line with animal welfare and dairy hygiene regulations. It will allow students to successfully grow and manage grass as a crop.

Programmes

AGRI-0092	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	2 / 4 / M

Indicative Content

- Dairy industry and milk production systems
- Milk composition and quality
- Feeding and management
- Replacement rearing
- Breeding management
- Housing and welfare
- Grass growth
- Grass management

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the structure of the dairy industry, milk marketing and milk production systems.
2. Discuss physiology of milk production, milk composition factors, milk quality best and quality standards.
3. Describe the feeding and management of a dairy herd.
4. Describe the feeding and management of replacement heifers and the principles of efficient breeding.
5. Describe all non-metabolic dairy cow diseases and disorders.
6. Discuss factors influencing grass growth, identify plant parts, grasses and weeds.
7. Describe grass management techniques, reseeding and grass conservation.

Learning and Teaching Methods

- Lectures.
- Practical instruction.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	2,3,5,7
Continuous Assessment	50%	
Assignment	50%	1,4,5,6

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter.

40%-49%: Will be able to show a basic knowledge of the key learning outcomes, including animal structures, functions and living processes as well as be able to perform simple laboratory experiments with animals.

50%-59%: As well as the above, the student will be able to demonstrate an understanding of some of the complexities of the topics, and predict nutrition requirements of the animal and potential nutritional disorders. The student will also be able to perform good quality experiments with animal tissues.

60%-69%: In addition, the student will demonstrate more detailed knowledge of the all topics, including and demonstrate an ability to evaluate detailed knowledge acquired in the module to new material. The student will also be able to perform very good quality experiments with animal tissues and microbiology related to animals.

70%-100%: The learner will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes, predict nutrition requirements of the animal and potential nutritional disorders and show the ability to evaluate and relate topics that overlap all of the learning outcomes. The student will also be able to perform high quality experiments in microbiology and with animal tissues.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Field Trips	12	
Independent Learning	87	

Essential Material

- Bell, P. _Dairy Herd Management_. Oakpark: Teagasc, 2001.
- Kennedy, J. __ _Breeidng__dairy cows_. Dublin: Irish Farmers Journal, 2011 .
- O' Dwyer , J. _Forage Production_. Oakpark: Teagasc, 2008.
- O' Dwyer, T. _Cash Cows_. Oakpark: Teagasc, 2011.
- O' Halloran, J. _Breeding and Calving Cows_. Oakpark: Teagasc, 2008.
- Teagasc 2011, A. _Teagasc__Dairy Manual. A best practise manual for Ireland's Dairy Farmers_. 1st ed. Oakpark: Teagasc, 2011.

Supplementary Material

- Chamberlain , A.T. _Feeding Dairy Cows_. London: Chalcombe, 1996.
- Throup , G. _Making Profits with Dairy Cows & Quotas_. London: Diamond Farm Book, 1998.